

# SOLIDWORKS API (.NET VBA C#) — Reference (239/337): IRenderMaterial (part 1 of 2)

SOLIDWORKS API Documentation (converted from CHM)

# IRenderMaterial (part 1 of 2) (Type)

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## IRenderMaterial Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRenderMaterial Interface

Use to apply appearances to models.

**NOTE:** In SOLIDWORKS 2008 and later, materials are called appearances. [RealView Graphics must be enabled](#) to see any applied appearances.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IRenderMaterial
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial
```

C#

```
public interface IRenderMaterial
```

C++/CLI

```
public interface class IRenderMaterial
```

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial](#).

### Example

[Add and Delete Appearances from Specific Display States \(C#\)](#)  
[Add and Delete Appearances from Specific Display States \(VB.NET\)](#)  
[Add and Delete Appearances from Specific Display States \(VBA\)](#)  
[Add Decal \(C#\)](#)  
[Add Decal \(VB.NET\)](#)  
[Add Decal \(VBA\)](#)

### Accessors

[IComponent2::GetRenderMaterials](#) and [IComponent2::IGetRenderMaterials](#)

[IModelDocExtension::CreateRenderMaterial](#)

[IModelDocExtension::GetRenderMaterials](#) and [IModelDocExtension::IGetRenderMaterials](#)

[ISwScene::GetFloorAppearance](#)

### Access Diagram

[RenderMaterial](#)

### See Also

[IRenderMaterial Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[IModelDocExtension::CreateRenderMaterial Method](#)  
[IModelDocExtension::GetRenderMaterialsCount Method](#)  
[IModelDocExtension::UpdateRenderMaterialsInSceneGraph Method](#)  
[IModelDocExtension::GetRenderCustomReferences Method](#)  
[IModelDocExtension::GetRenderStockReferences Method](#)

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# IRenderMaterial Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRenderMaterial Interface

For a list of all members of this type, see [IRenderMaterial members](#).

## Public Methods

| Name                                                                                                                       | Description                                                             |
|----------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------|
|  <a href="#">AddEntity</a>                | Adds the appearance to the specified entity.                            |
|  <a href="#">GetCenterPoint</a>           | Obsolete. Superseded by <a href="#">IRenderMaterial.GetCenterPoint2</a> |
|  <a href="#">GetCenterPoint2</a>          | Gets the center point of the mapping for texture-based appearances.     |
|  <a href="#">GetDSPBRMaterial</a>         | Gets the DSPBRMaterial object.                                          |
|  <a href="#">GetEntities</a>              | Gets the entities on which this appearance is applied.                  |
|  <a href="#">GetEntitiesCount</a>         | Gets the number of entities on which this appearance was applied.       |
|  <a href="#">GetLinkedDisplayStates</a>   | Gets the display states to which this appearance is applied.            |
|  <a href="#">GetMaterialIds</a>           | Gets the material IDs of an appearance.                                 |
|  <a href="#">GetUDirection</a>            | Obsolete. Superseded by <a href="#">IRenderMaterial.GetUDirection2</a>  |
|  <a href="#">GetUDirection2</a>           | Gets the U direction of the texture-based appearance.                   |
|  <a href="#">GetVDirection</a>            | Obsolete. Superseded by <a href="#">IRenderMaterial.GetVDirection2</a>  |
|  <a href="#">GetVDirection2</a>           | Gets the V direction of the texture-based appearance.                   |
|  <a href="#">IGetEntities</a>             | Gets the entities on which this appearance is applied.                  |
|  <a href="#">RemoveAllEntities</a>        | Removes the appearance from all entities on which it is applied.        |
|  <a href="#">SetCenterPoint</a>           | Obsolete. Superseded by <a href="#">IRenderMaterial.SetCenterPoint2</a> |
|  <a href="#">SetCenterPoint2</a>         | Sets the center point of the mapping for texture-based appearances.     |
|  <a href="#">SetLinkedDisplayStates</a> | Sets the display states to which this appearance is applied.            |
|  <a href="#">SetUDirection</a>          | Obsolete. Superseded by <a href="#">IRenderMaterial.SetUDirection2</a>  |
|  <a href="#">SetUDirection2</a>         | Sets the U direction of the texture-based appearance.                   |
|  <a href="#">SetVDirection</a>          | Obsolete. Superseded by <a href="#">IRenderMaterial.SetVDirection2</a>  |
|  <a href="#">SetVDirection2</a>         | Sets the V direction of the texture-based appearance.                   |

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## See Also

- [IRenderMaterial Interface](#)
- [SolidWorks.Interop.sldworks Namespace](#)
- [IModelDocExtension.CreateRenderMaterial Method](#)
- [IModelDocExtension.GetRenderMaterialsCount Method](#)
- [IModelDocExtension.UpdateRenderMaterialsInSceneGraph Method](#)
- [IModelDocExtension.GetRenderCustomReferences Method](#)
- [IModelDocExtension.GetRenderStockReferences Method](#)

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# IRenderMaterial Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IRenderMaterial Interface

The following tables list the members exposed by [IRenderMaterial](#).

## Public Properties

| Name                                                                                                                    | Description                                                                                                                                            |
|-------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">AboutAxis</a>           | Gets or sets the about axis value for a cylindrical mapping type.                                                                                      |
|  <a href="#">AccurateReflections</a> | Selects or clears the <b>Accurate reflections (slower)</b> setting, which controls the level of surface reflections, for illuminating this appearance. |

|                                                                                                                               |                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">AlongAxis</a>                   | Gets or sets the along axis value for a cylindrical mapping type.                                                                                                                                |
|  <a href="#">Ambient</a>                     | Gets or sets the ambient light intensity for illuminating this appearance.                                                                                                                       |
|  <a href="#">AnisotropicBias</a>             | Gets or sets the anisotropic bias, which makes the effect of light on the surface stronger in the horizontal or vertical direction, for illuminating this appearance.                            |
|  <a href="#">AnisotropicCylinderDistance</a> | Gets or sets the anisotropic cylinder distance for illuminating this appearance.                                                                                                                 |
|  <a href="#">AnisotropicFloorHeight</a>      | Gets or sets the anisotropic floor height for illuminating this appearance.                                                                                                                      |
|  <a href="#">Brightness</a>                  | Gets or sets how emissive the material is for the Constant illumination type for this appearance.                                                                                                |
|  <a href="#">BumpAmplitude</a>               | Obsolete. Superseded by <a href="#">IRenderMaterial.BumpStrength</a> .                                                                                                                           |
|  <a href="#">BumpBlend</a>                   | Gets or sets the blend, which is the extent of the boundary between each bump and the surface, of the surface finish for this appearance.                                                        |
|  <a href="#">BumpDetail</a>                  | Gets or sets the level of granularity for any surface finish for this appearance.                                                                                                                |
|  <a href="#">BumpMap</a>                     | Obsolete. Superseded by <a href="#">IRenderMaterial.SurfaceFinishShaderType</a> .                                                                                                                |
|  <a href="#">BumpRadius</a>                  | Gets or sets the radius, which controls the relative size and spacing of bumps for dimpled and tread plate styles, of the surface finish for this appearance.                                    |
|  <a href="#">BumpRoughHigh</a>               | Gets or sets the high threshold of the surface finish for this appearance.                                                                                                                       |
|  <a href="#">BumpRoughLow</a>                | Gets or sets the low threshold of the surface finish for this appearance.                                                                                                                        |
|  <a href="#">BumpScale</a>                   | Gets or sets the scale for the surface-finish pattern incidence for this appearance.                                                                                                             |
|  <a href="#">BumpSharpness</a>               | Gets or sets the sharpness, which influences the shape of the surface finish, of this appearance.                                                                                                |
|  <a href="#">BumpStrength</a>                | Gets or sets the amplitude of the surface layer for this appearance.                                                                                                                             |
|  <a href="#">BumpTextureFilename</a>         | Obsolete. Superseded by <a href="#">IRenderMaterial.SurfaceFinishFile</a> . Gets or sets the path and file name of the pattern based on an image file for the surface finish of this appearance. |
|  <a href="#">BumpUseMappingScale</a>         | Gets or sets whether to use the material's scale and mapping for the surface finish of this appearance.                                                                                          |
|  <a href="#">CausticsCast</a>                | Gets or sets whether specular materials reflect caustic photons for illuminating this appearance.                                                                                                |
|  <a href="#">CausticsReceive</a>             | Gets or sets whether diffuse materials absorb caustic photons for illuminating this appearance.                                                                                                  |
|  <a href="#">ColorForm</a>                   | Gets or sets the number of colors required to describe the appearance .                                                                                                                          |
|  <a href="#">CustomHoleMeshFile</a>         | Gets or sets the custom hole mesh file.                                                                                                                                                          |
|  <a href="#">DensityOfHoles</a>            | Gets or sets the density of the holes of the mesh in corroded or eroded materials for illuminating this appearance.                                                                              |
|  <a href="#">Diffuse</a>                   | Gets or sets the intensity of a light source illuminating a surface from all directions without attenuation or shadowing for this appearance.                                                    |
|  <a href="#">DimpleSize</a>                | Gets or sets the dimple size.                                                                                                                                                                    |
|  <a href="#">Direction1RotationAngle</a>   | Gets or sets the angle at which to rotate the texture relative to the horizontal axis for mapping this appearance.                                                                               |
|  <a href="#">Direction2RotationAngle</a>   | Gets or sets the angle at which to rotate the texture relative to the vertical axis for mapping this appearance.                                                                                 |
|  <a href="#">DoubleSided</a>               | Gets or sets whether to enable shading from both sides of a surface when rendering a model using a ray-trace rendering engine.                                                                   |
|  <a href="#">Emission</a>                  | Gets or sets how much light is projected from the appearance.                                                                                                                                    |
|  <a href="#">FaceWidth</a>                 | Gets or sets the face width.                                                                                                                                                                     |
|  <a href="#">FileName</a>                  | Gets or sets the path and file name (.p2m) of the appearance.                                                                                                                                    |
|  <a href="#">FitHeight</a>                 | Gets or sets whether to stretch the height of the appearance texture to the selected entity when mapping the appearance.                                                                         |
|  <a href="#">FitWidth</a>                  | Gets or sets whether to stretch the width of the appearance texture to the selected entity when mapping the appearance.                                                                          |
|  <a href="#">FixedAspectRatio</a>          | Gets or sets whether the aspect ratio is fixed for mapping this appearance texture.                                                                                                              |
|  <a href="#">GlobalIlluminationCast</a>    | Gets whether specular materials reflect photons for illuminating this appearance..                                                                                                               |
|  <a href="#">GlobalIlluminationReceive</a> | Gets or sets whether diffuse materials absorb photons for illuminating this appearance.                                                                                                          |
|  <a href="#">Glossy</a>                    | Gets or sets the specular factor of the lights reflected by the appearance.                                                                                                                      |
|  <a href="#">HasDSPBRProperties</a>        | Gets whether this appearance has DSPBR properties.                                                                                                                                               |
|  <a href="#">Height</a>                    | Gets or sets the height for mapping this appearance texture.                                                                                                                                     |
|  <a href="#">HeightMirror</a>              | Gets or sets whether to flip the appearance texture about the selected direction vertically.                                                                                                     |
|  <a href="#">HoleMeshSize</a>              | Gets or sets the hole mesh size.                                                                                                                                                                 |
|  <a href="#">IgnoreMissingFile</a>         | Gets or sets whether to ignore any missing image file warnings.                                                                                                                                  |
| <a href="#">IlluminationShaderType</a>                                                                                        | Gets or sets the type of illumination of the appearance.                                                                                                                                         |

|  |                                            |                                                                                                                                |
|--|--------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
|  |                                            |                                                                                                                                |
|  | <a href="#">IndexOfRefraction</a>          | Gets or sets the index of refraction for illuminating this appearance.                                                         |
|  | <a href="#">KnurlAngle</a>                 | Gets or sets the knurl angle of this appearance.                                                                               |
|  | <a href="#">KnurlHeightCap</a>             | Gets or sets the maximum knurl height of this appearance.                                                                      |
|  | <a href="#">LinkToFile</a>                 | Gets or sets whether to link instances of the appearance to an appearance file (.p2m).                                         |
|  | <a href="#">MappingType</a>                | Gets or sets the mapping type for this appearance.                                                                             |
|  | <a href="#">MaterialID</a>                 | <b>Not supported in SOLIDWORKS 2011 and later. Superseded by <a href="#">JRenderMaterial.GetMaterialIds</a></b>                |
|  | <a href="#">MetallicAmplitude</a>          | Gets or sets the amplitude of the metallic flakes for illuminating the appearance.                                             |
|  | <a href="#">MetallicFlakeMaterial</a>      | Gets or sets the metallic flake material for illuminating the appearance.                                                      |
|  | <a href="#">MetallicMix</a>                | Gets or sets the metallic quality of a material for illuminating the appearance.                                               |
|  | <a href="#">MetallicRoughness</a>          | Gets or sets the size (coarseness) any highlights on the surface for illuminating the appearance.                              |
|  | <a href="#">MetallicScale</a>              | Gets or sets the size of the metallic flakes in the metallic layer for illuminating the appearance.                            |
|  | <a href="#">NSamples</a>                   | Gets or sets the number of samples used to calculate the contribution of the glossy component for illuminating the appearance. |
|  | <a href="#">ObjectAreaLight</a>            | Gets or sets whether the appearance is an object area light or whether it has realistic falloff, or both.                      |
|  | <a href="#">OffsetLatitude</a>             | Gets or sets the offset latitude value for a spherical mapping type.                                                           |
|  | <a href="#">OffsetLongitude</a>            | Gets or sets the offset longitude value for a spherical mapping type.                                                          |
|  | <a href="#">PatternScale</a>               | Gets or sets the pattern scale of procedural textures for mapping the appearance.                                              |
|  | <a href="#">PrimaryColor</a>               | Gets or sets the primary color of the appearance.                                                                              |
|  | <a href="#">ProjectionReference</a>        | Gets or sets the projection direction for mapping the appearance.                                                              |
|  | <a href="#">Reflectivity</a>               | Gets or sets the reflectivity for illuminating the appearance.                                                                 |
|  | <a href="#">RotationAngle</a>              | Gets or sets the angle by which to rotate the texture relative to the axes for mapping this appearance.                        |
|  | <a href="#">Roughness</a>                  | Gets or sets the specular spread (roughness) of the appearance.                                                                |
|  | <a href="#">RoundSharpEdges</a>            | Gets or sets the value by which to round sharp edges when rendering a model using a ray-trace rendering engine.                |
|  | <a href="#">SecondaryColor</a>             | Gets or sets the secondary color of the appearance.                                                                            |
|  | <a href="#">Specular</a>                   | Gets or sets the intensity of the light surface for illuminating the appearance.                                               |
|  | <a href="#">SpecularColor</a>              | Gets or sets the specular color for illuminating this appearance.                                                              |
|  | <a href="#">SurfaceFinishFile</a>          | Gets or sets the path and file name of the pattern based on an image file for the surface finish of this appearance.           |
|  | <a href="#">SurfaceFinishFileNormalMap</a> | Gets or sets whether to map the surface finish file image to the normal.                                                       |
|  | <a href="#">SurfaceFinishShaderType</a>    | Gets or sets the type of surface finish for the appearance.                                                                    |
|  | <a href="#">TertiaryColor</a>              | Gets or sets the tertiary color of the appearance.                                                                             |
|  | <a href="#">TextureFilename</a>            | Gets or sets the path and filename of the texture for this appearance.                                                         |
|  | <a href="#">ToonShaderTextureFilename</a>  | Gets or sets the path and filename for the toon shader texture file.                                                           |
|  | <a href="#">Translucency</a>               | Gets or sets the degree to which the material is able to filter and diffuse light for illuminating the appearance.             |
|  | <a href="#">Transparency</a>               | Gets or sets the degree to which a material allows light to pass through for illuminating the appearance.                      |
|  | <a href="#">Width</a>                      | Gets or sets the width for mapping this appearance texture.                                                                    |
|  | <a href="#">WidthMirror</a>                | Gets or sets whether to flip the appearance texture about the selected direction horizontally.                                 |
|  | <a href="#">XPosition</a>                  | Gets or sets the X offset direction for the appearance.                                                                        |
|  | <a href="#">YPosition</a>                  | Gets or sets the Y offset direction for the appearance.                                                                        |

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## Public Methods

| Name                      | Description                                  |
|---------------------------|----------------------------------------------|
| <a href="#">AddEntity</a> | Adds the appearance to the specified entity. |

|                                                                                   |                                        |                                                                         |
|-----------------------------------------------------------------------------------|----------------------------------------|-------------------------------------------------------------------------|
|  | <a href="#">GetCenterPoint</a>         | Obsolete. Superseded by <a href="#">IRenderMaterial.GetCenterPoint2</a> |
|  | <a href="#">GetCenterPoint2</a>        | Gets the center point of the mapping for texture-based appearances.     |
|  | <a href="#">GetDSPBRMaterial</a>       | Gets the DSPBRMaterial object.                                          |
|  | <a href="#">GetEntities</a>            | Gets the entities on which this appearance is applied.                  |
|  | <a href="#">GetEntitiesCount</a>       | Gets the number of entities on which this appearance was applied.       |
|  | <a href="#">GetLinkedDisplayStates</a> | Gets the display states to which this appearance is applied.            |
|  | <a href="#">GetMaterialIds</a>         | Gets the material IDs of an appearance.                                 |
|  | <a href="#">GetUDirection</a>          | Obsolete. Superseded by <a href="#">IRenderMaterial.GetUDirection2</a>  |
|  | <a href="#">GetUDirection2</a>         | Gets the U direction of the texture-based appearance.                   |
|  | <a href="#">GetVDirection</a>          | Obsolete. Superseded by <a href="#">IRenderMaterial.GetVDirection2</a>  |
|  | <a href="#">GetVDirection2</a>         | Gets the V direction of the texture-based appearance.                   |
|  | <a href="#">IGetEntities</a>           | Gets the entities on which this appearance is applied.                  |
|  | <a href="#">RemoveAllEntities</a>      | Removes the appearance from all entities on which it is applied.        |
|  | <a href="#">SetCenterPoint</a>         | Obsolete. Superseded by <a href="#">IRenderMaterial.SetCenterPoint2</a> |
|  | <a href="#">SetCenterPoint2</a>        | Sets the center point of the mapping for texture-based appearances.     |
|  | <a href="#">SetLinkedDisplayStates</a> | Sets the display states to which this appearance is applied.            |
|  | <a href="#">SetUDirection</a>          | Obsolete. Superseded by <a href="#">IRenderMaterial.SetUDirection2</a>  |
|  | <a href="#">SetUDirection2</a>         | Sets the U direction of the texture-based appearance.                   |
|  | <a href="#">SetVDirection</a>          | Obsolete. Superseded by <a href="#">IRenderMaterial.SetVDirection2</a>  |
|  | <a href="#">SetVDirection2</a>         | Sets the V direction of the texture-based appearance.                   |

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## See Also

[IRenderMaterial Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[IModelDocExtension::CreateRenderMaterial Method](#)  
[IModelDocExtension::GetRenderMaterialsCount Method](#)  
[IModelDocExtension::UpdateRenderMaterialsInSceneGraph Method](#)  
[IModelDocExtension::GetRenderCustomReferences Method](#)  
[IModelDocExtension::GetRenderStockReferences Method](#)

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## IRenderMaterial Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IRenderMaterial Interface

For a list of all members of this type, see [IRenderMaterial members](#).

## Public Properties

| Name                                                                                                                            | Description                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">AboutAxis</a>                   | Gets or sets the about axis value for a cylindrical mapping type.                                                                                                     |
|  <a href="#">AccurateReflections</a>         | Selects or clears the <b>Accurate reflections (slower)</b> setting, which controls the level of surface reflections, for illuminating this appearance.                |
|  <a href="#">AlongAxis</a>                   | Gets or sets the along axis value for a cylindrical mapping type.                                                                                                     |
|  <a href="#">Ambient</a>                     | Gets or sets the ambient light intensity for illuminating this appearance.                                                                                            |
|  <a href="#">AnisotropicBias</a>             | Gets or sets the anisotropic bias, which makes the effect of light on the surface stronger in the horizontal or vertical direction, for illuminating this appearance. |
|  <a href="#">AnisotropicCylinderDistance</a> | Gets or sets the anisotropic cylinder distance for illuminating this appearance.                                                                                      |
|  <a href="#">AnisotropicFloorHeight</a>      | Gets or sets the anisotropic floor height for illuminating this appearance.                                                                                           |
|  <a href="#">Brightness</a>                  | Gets or sets how emissive the material is for the Constant illumination type for this appearance.                                                                     |
|  <a href="#">BumpAmplitude</a>               | Obsolete. Superseded by <a href="#">IRenderMaterial.BumpStrength</a> .                                                                                                |
| <a href="#">BumpBlend</a>                                                                                                       | Gets or sets the blend, which is the extent of the boundary between each bump and the surface, of the surface finish for this appearance.                             |

|                                                                                     |                                           |                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|     |                                           |                                                                                                                                                                                                  |
|    | <a href="#">BumpDetail</a>                | Gets or sets the level of granularity for any surface finish for this appearance.                                                                                                                |
|    | <a href="#">BumpMap</a>                   | Obsolete. Superseded by <a href="#">IRenderMaterial.SurfaceFinishShaderType</a> .                                                                                                                |
|    | <a href="#">BumpRadius</a>                | Gets or sets the radius, which controls the relative size and spacing of bumps for dimpled and tread plate styles, of the surface finish for this appearance.                                    |
|    | <a href="#">BumpRoughHigh</a>             | Gets or sets the high threshold of the surface finish for this appearance.                                                                                                                       |
|    | <a href="#">BumpRoughLow</a>              | Gets or sets the low threshold of the surface finish for this appearance.                                                                                                                        |
|    | <a href="#">BumpScale</a>                 | Gets or sets the scale for the surface-finish pattern incidence for this appearance.                                                                                                             |
|    | <a href="#">BumpSharpness</a>             | Gets or sets the sharpness, which influences the shape of the surface finish, of this appearance.                                                                                                |
|    | <a href="#">BumpStrength</a>              | Gets or sets the amplitude of the surface layer for this appearance.                                                                                                                             |
|    | <a href="#">BumpTextureFilename</a>       | Obsolete. Superseded by <a href="#">IRenderMaterial.SurfaceFinishFile</a> . Gets or sets the path and file name of the pattern based on an image file for the surface finish of this appearance. |
|    | <a href="#">BumpUseMappingScale</a>       | Gets or sets whether to use the material's scale and mapping for the surface finish of this appearance.                                                                                          |
|    | <a href="#">CausticsCast</a>              | Gets or sets whether specular materials reflect caustic photons for illuminating this appearance.                                                                                                |
|    | <a href="#">CausticsReceive</a>           | Gets or sets whether diffuse materials absorb caustic photons for illuminating this appearance.                                                                                                  |
|    | <a href="#">ColorForm</a>                 | Gets or sets the number of colors required to describe the appearance .                                                                                                                          |
|    | <a href="#">CustomHoleMeshFile</a>        | Gets or sets the custom hole mesh file.                                                                                                                                                          |
|    | <a href="#">DensityOfHoles</a>            | Gets or sets the density of the holes of the mesh in corroded or eroded materials for illuminating this appearance.                                                                              |
|    | <a href="#">Diffuse</a>                   | Gets or sets the intensity of a light source illuminating a surface from all directions without attenuation or shadowing for this appearance.                                                    |
|    | <a href="#">DimpleSize</a>                | Gets or sets the dimple size.                                                                                                                                                                    |
|    | <a href="#">Direction1RotationAngle</a>   | Gets or sets the angle at which to rotate the texture relative to the horizontal axis for mapping this appearance.                                                                               |
|    | <a href="#">Direction2RotationAngle</a>   | Gets or sets the angle at which to rotate the texture relative to the vertical axis for mapping this appearance.                                                                                 |
|    | <a href="#">DoubleSided</a>               | Gets or sets whether to enable shading from both sides of a surface when rendering a model using a ray-trace rendering engine.                                                                   |
|   | <a href="#">Emission</a>                  | Gets or sets how much light is projected from the appearance.                                                                                                                                    |
|  | <a href="#">FaceWidth</a>                 | Gets or sets the face width.                                                                                                                                                                     |
|  | <a href="#">FileName</a>                  | Gets or sets the path and file name ( <b>.p2m</b> ) of the appearance.                                                                                                                           |
|  | <a href="#">FitHeight</a>                 | Gets or sets whether to stretch the height of the appearance texture to the selected entity when mapping the appearance.                                                                         |
|  | <a href="#">FitWidth</a>                  | Gets or sets whether to stretch the width of the appearance texture to the selected entity when mapping the appearance.                                                                          |
|  | <a href="#">FixedAspectRatio</a>          | Gets or sets whether the aspect ratio is fixed for mapping this appearance texture.                                                                                                              |
|  | <a href="#">GlobalIlluminationCast</a>    | Gets whether specular materials reflect photons for illuminating this appearance..                                                                                                               |
|  | <a href="#">GlobalIlluminationReceive</a> | Gets or sets whether diffuse materials absorb photons for illuminating this appearance.                                                                                                          |
|  | <a href="#">Glossy</a>                    | Gets or sets the specular factor of the lights reflected by the appearance.                                                                                                                      |
|  | <a href="#">HasDSPBRProperties</a>        | Gets whether this appearance has DSPBR properties.                                                                                                                                               |
|  | <a href="#">Height</a>                    | Gets or sets the height for mapping this appearance texture.                                                                                                                                     |
|  | <a href="#">HeightMirror</a>              | Gets or sets whether to flip the appearance texture about the selected direction vertically.                                                                                                     |
|  | <a href="#">HoleMeshSize</a>              | Gets or sets the hole mesh size.                                                                                                                                                                 |
|  | <a href="#">IgnoreMissingFile</a>         | Gets or sets whether to ignore any missing image file warnings.                                                                                                                                  |
|  | <a href="#">IlluminationShaderType</a>    | Gets or sets the type of illumination of the appearance.                                                                                                                                         |
|  | <a href="#">IndexOfRefraction</a>         | Gets or sets the index of refraction for illuminating this appearance.                                                                                                                           |
|  | <a href="#">KnurlAngle</a>                | Gets or sets the knurl angle of this appearance.                                                                                                                                                 |
|  | <a href="#">KnurlHeightCap</a>            | Gets or sets the maximum knurl height of this appearance.                                                                                                                                        |
|  | <a href="#">LinkToFile</a>                | Gets or sets whether to link instances of the appearance to an appearance file ( <b>.p2m</b> ).                                                                                                  |
|  | <a href="#">MappingType</a>               | Gets or sets the mapping type for this appearance.                                                                                                                                               |
|  | <a href="#">MaterialID</a>                | Not supported in SOLIDWORKS 2011 and later. Superseded by <a href="#">IRenderMaterial.GetMaterialIds</a> .                                                                                       |
|                                                                                     | <a href="#">MetallicAmplitude</a>         | Gets or sets the amplitude of the metallic flakes for illuminating the appearance.                                                                                                               |

|                                                                                                                                |                                                                                                                                |  |
|--------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|--|
|                                                |                                                                                                                                |  |
|  <a href="#">MetallicFlakeMaterial</a>        | Gets or sets the metallic flake material for illuminating the appearance.                                                      |  |
|  <a href="#">MetallicMix</a>                  | Gets or sets the metallic quality of a material for illuminating the appearance.                                               |  |
|  <a href="#">MetallicRoughness</a>            | Gets or sets the size (coarseness) any highlights on the surface for illuminating the appearance.                              |  |
|  <a href="#">MetallicScale</a>                | Gets or sets the size of the metallic flakes in the metallic layer for illuminating the appearance.                            |  |
|  <a href="#">NSamples</a>                     | Gets or sets the number of samples used to calculate the contribution of the glossy component for illuminating the appearance. |  |
|  <a href="#">ObjectAreaLight</a>              | Gets or sets whether the appearance is an object area light or whether it has realistic falloff, or both.                      |  |
|  <a href="#">OffsetLatitude</a>               | Gets or sets the offset latitude value for a spherical mapping type.                                                           |  |
|  <a href="#">OffsetLongitude</a>              | Gets or sets the offset longitude value for a spherical mapping type.                                                          |  |
|  <a href="#">PatternScale</a>                 | Gets or sets the pattern scale of procedural textures for mapping the appearance.                                              |  |
|  <a href="#">PrimaryColor</a>                 | Gets or sets the primary color of the appearance.                                                                              |  |
|  <a href="#">ProjectionReference</a>          | Gets or sets the projection direction for mapping the appearance.                                                              |  |
|  <a href="#">Reflectivity</a>                 | Gets or sets the reflectivity for illuminating the appearance.                                                                 |  |
|  <a href="#">RotationAngle</a>                | Gets or sets the angle by which to rotate the texture relative to the axes for mapping this appearance.                        |  |
|  <a href="#">Roughness</a>                    | Gets or sets the specular spread (roughness) of the appearance.                                                                |  |
|  <a href="#">RoundSharpEdges</a>              | Gets or sets the value by which to round sharp edges when rendering a model using a ray-trace rendering engine.                |  |
|  <a href="#">SecondaryColor</a>               | Gets or sets the secondary color of the appearance.                                                                            |  |
|  <a href="#">Specular</a>                     | Gets or sets the intensity of the light surface for illuminating the appearance.                                               |  |
|  <a href="#">SpecularColor</a>                | Gets or sets the specular color for illuminating this appearance.                                                              |  |
|  <a href="#">SurfaceFinishFile</a>           | Gets or sets the path and file name of the pattern based on an image file for the surface finish of this appearance.           |  |
|  <a href="#">SurfaceFinishFileNormalMap</a> | Gets or sets whether to map the surface finish file image to the normal.                                                       |  |
|  <a href="#">SurfaceFinishShaderType</a>    | Gets or sets the type of surface finish for the appearance.                                                                    |  |
|  <a href="#">TertiaryColor</a>              | Gets or sets the tertiary color of the appearance.                                                                             |  |
|  <a href="#">TextureFilename</a>            | Gets or sets the path and filename of the texture for this appearance.                                                         |  |
|  <a href="#">ToonShaderTextureFilename</a>  | Gets or sets the path and filename for the toon shader texture file.                                                           |  |
|  <a href="#">Translucency</a>               | Gets or sets the degree to which the material is able to filter and diffuse light for illuminating the appearance.             |  |
|  <a href="#">Transparency</a>               | Gets or sets the degree to which a material allows light to pass through for illuminating the appearance.                      |  |
|  <a href="#">Width</a>                      | Gets or sets the width for mapping this appearance texture.                                                                    |  |
|  <a href="#">WidthMirror</a>                | Gets or sets whether to flip the appearance texture about the selected direction horizontally.                                 |  |
|  <a href="#">XPosition</a>                  | Gets or sets the X offset direction for the appearance.                                                                        |  |
|  <a href="#">YPosition</a>                  | Gets or sets the Y offset direction for the appearance.                                                                        |  |

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 See Also

- [IRenderMaterial Interface](#)
- [SolidWorks.Interop.sldworks Namespace](#)
- [IModelDocExtension::CreateRenderMaterial Method](#)
- [IModelDocExtension::GetRenderMaterialsCount Method](#)
- [IModelDocExtension::UpdateRenderMaterialsInSceneGraph Method](#)
- [IModelDocExtension::GetRenderCustomReferences Method](#)
- [IModelDocExtension::GetRenderStockReferences Method](#)

AboutAxis Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AboutAxis Property (IRenderMaterial)

Gets or sets the about axis value for a cylindrical mapping type.

## Visual Basic Syntax

Visual Basic (Declaration)

Property AboutAxis As System.Double

Visual Basic (Usage)



```
Dim instance As IRenderMaterial
Dim value As System.Double
```

```
instance.AboutAxis = value
```

```
value = instance.AboutAxis
```

C#

```
System.double AboutAxis {get; set;}
```

C++/CLI

```
property System.double AboutAxis {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

About axis value

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AboutAxis](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2025 FCS, Revision Number 33

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## AccurateReflections Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AccurateReflections Property (IRenderMaterial)

Selects or clears the **Accurate reflections (slower)** setting, which controls the level of surface reflections, for illuminating this appearance.

## Visual Basic Syntax

Visual Basic (Declaration)

Property AccurateReflections As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRenderMaterial
Dim value As System.Boolean
```

```
instance.AccurateReflections = value
```

```
value = instance.AccurateReflections
```

C#

```
System.bool AccurateReflections {get; set;}
```

C++/CLI

```
property System.bool AccurateReflections {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to select the **Accurate reflections (slower)** setting, false to clear it

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AccurateReflections](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## AddEntity Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AddEntity Method (IRenderMaterial)

Adds the appearance to the specified entity.

### .NET Syntax

Visual Basic (Declaration)

```
Function AddEntity(  
    ByVal Entity As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim Entity As System.Object  
Dim value As System.Boolean
```

```
value = instance.AddEntity(Entity)
```

C#

```
System.bool AddEntity(  
    System.object Entity  
)
```

C++/CLI

```
System.bool AddEntity(  
& System.Object^ Entity  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*Entity*  
Entity to which to add the appearance

## Return Value

True if the appearance is added to Entity, false if not

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AddEntity](#).

## Example

See [IRenderMaterial](#) examples.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::GetEntities Method](#)  
[IRenderMaterial::GetEntitiesCount Method](#)  
[IRenderMaterial::IGetEntities Method](#)  
[IRenderMaterial::RemoveAllEntities Method](#)

## Availability

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## AlongAxis Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AlongAxis Property (IRenderMaterial)

Gets or sets the along axis value for a cylindrical mapping type.

## .NET Syntax

Visual Basic (Declaration)

Property AlongAxis As System.Double

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double  
  
instance.AlongAxis = value  
  
value = instance.AlongAxis
```

C#

```
System.double AlongAxis {get; set;}
```

C++/CLI

```
property System.double AlongAxis {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Along axis value

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AlongAxis](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## Ambient Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : Ambient Property (IRenderMaterial)

Gets or sets the ambient light intensity for illuminating this appearance.

## .NET Syntax

### Visual Basic (Declaration)

```
Property Ambient As System.Double
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.Ambient = value
```

```
value = instance.Ambient
```

### C#

```
System.double Ambient {get; set;}
```

### C++/CLI

```
property System.double Ambient {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Ambient light intensity

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::Ambient](#).

## Remarks

The scene should contain an ambient light.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## AnisotropicBias Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AnisotropicBias Property (IRenderMaterial)

Gets or sets the anisotropic bias, which makes the effect of light on the surface stronger in the horizontal or vertical direction, for illuminating this appearance.

## .NET Syntax

### Visual Basic (Declaration)

```
Property AnisotropicBias As System.Double
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.AnisotropicBias = value
```

```
value = instance.AnisotropicBias
```

### C#

```
System.double AnisotropicBias {get; set;}
```

### C++/CLI

```
property System.double AnisotropicBias {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Anisotropic bias

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AnisotropicBias](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## AnisotropicCylinderDistance Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AnisotropicCylinderDistance Property (IRenderMaterial)

Gets or sets the anisotropic cylinder distance for illuminating this appearance.

## .NET Syntax

### Visual Basic (Declaration)

```
Property AnisotropicCylinderDistance As System.Double
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.AnisotropicCylinderDistance = value
```

```
value = instance.AnisotropicCylinderDistance
```

C#

```
System.double AnisotropicCylinderDistance {get; set;}
```

C++/CLI

```
property System.double AnisotropicCylinderDistance {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Anisotropic cylinder distance

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AnisotropicCylinderDistance](#).

### Remarks

The anisotropic reflection is modeled by laying (virtual) cylinders along the surface. This property controls the distance between the cylinders. For a stronger anisotropic effect, increase the cylinder distance.

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## AnisotropicFloorHeight Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : AnisotropicFloorHeight Property (IRenderMaterial)

Gets or sets the anisotropic floor height for illuminating this appearance.

### .NET Syntax

Visual Basic (Declaration)

```
Property AnisotropicFloorHeight As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double  
  
instance.AnisotropicFloorHeight = value  
  
value = instance.AnisotropicFloorHeight
```

C#

```
System.double AnisotropicFloorHeight {get; set;}
```

C++/CLI

```
property System.double AnisotropicFloorHeight {  
    System.double get();
```

```
void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Anisotropic floor height

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::AnisotropicFloorHeight](#).

### Remarks

Anisotropic reflection is modeled by laying (virtual) cylinders along the surface. This property controls the height difference between neighboring cylinders. For a stronger anisotropic effect, decrease the floor height.

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## Brightness Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : Brightness Property (IRenderMaterial)

Gets or sets how emissive the material is for the Constant illumination type for this appearance.

### .NET Syntax

Visual Basic (Declaration)

```
Property Brightness As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.Brightness = value
```

```
value = instance.Brightness
```

C#

```
System.double Brightness {get; set;}
```

C++/CLI

```
property System.double Brightness {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Brightness

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::Brightness](#).

## Remarks

This property corresponds to the Brightness control on the Illumination tab of the Appearances PropertyManager page when Illumination is Constant.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## BumpAmplitude Property (IRenderMaterial)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > [IRenderMaterial Interface](#) : BumpAmplitude Property (IRenderMaterial)

Obsolete. Superseded by [IRenderMaterial.BumpStrength](#).

## .NET Syntax

### Visual Basic (Declaration)

Property BumpAmplitude As System.Double

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpAmplitude = value
```

```
value = instance.BumpAmplitude
```

### C#

```
System.double BumpAmplitude {get; set;}
```

### C++/CLI

```
property System.double BumpAmplitude {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Amplitude

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.BumpAmplitude](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## BumpBlend Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpBlend Property (IRenderMaterial)

Gets or sets the blend, which is the extent of the boundary between each bump and the surface, of the surface finish for this appearance.

## .NET Syntax

### Visual Basic (Declaration)

```
Property BumpBlend As System.Double
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpBlend = value
```

```
value = instance.BumpBlend
```

### C#

```
System.double BumpBlend {get; set;}
```

### C++/CLI

```
property System.double BumpBlend {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Extent of the boundary between each bump and the surface

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.BumpBlend](#).

## Remarks

This property supports dimpled, tread plate, and knurled styles of surface finishes.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## BumpDetail Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpDetail Property (IRenderMaterial)

Gets or sets the level of granularity for any surface finish for this appearance.

## .NET Syntax

### Visual Basic (Declaration)

```
Property BumpDetail As System.Integer
```

### Visual Basic (Usage)

```
Dim instance As IRenderMaterial
Dim value As System.Integer

instance.BumpDetail = value

value = instance.BumpDetail
```

C#

```
System.int BumpDetail {get; set;}
```

C++/CLI

```
property System.int BumpDetail {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Level of granularity

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpDetail](#).

### Remarks

This property is only available when Casting, Rough, Chips, Circular or Rough/Smooth is selected for the [surface finish](#). When set to high detail, individual surface elements appear in sharp focus; when set to low detail, surface elements appear in soft focus.

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## BumpMap Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpMap Property (IRenderMaterial)

**Obsolete.** Superseded by [IRenderMaterial::SurfaceFinishShaderType](#).

### .NET Syntax

Visual Basic (Declaration)

```
Property BumpMap As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IRenderMaterial
Dim value As System.Integer

instance.BumpMap = value

value = instance.BumpMap
```

C#

```
System.int BumpMap {get; set;}
```

C++/CLI

```
property System.int BumpMap {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Surface finish as defined by [swRenderMaterialBumpMap\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpMap](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## BumpRadius Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpRadius Property (IRenderMaterial)

Gets or sets the radius, which controls the relative size and spacing of bumps for dimpled and tread plate styles, of the surface finish for this appearance.

### .NET Syntax

Visual Basic (Declaration)

```
Property BumpRadius As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpRadius = value
```

```
value = instance.BumpRadius
```

C#

```
System.double BumpRadius {get; set;}
```

C++/CLI

```
property System.double BumpRadius {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Radius

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpRadius](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

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## BumpRoughHigh Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpRoughHigh Property (IRenderMaterial)

Gets or sets the high threshold of the surface finish for this appearance.

## .NET Syntax

Visual Basic (Declaration)

Property BumpRoughHigh As System.Double

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpRoughHigh = value
```

```
value = instance.BumpRoughHigh
```

C#

```
System.double BumpRoughHigh {get; set;}
```

C++/CLI

```
property System.double BumpRoughHigh {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

High threshold

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpRoughHigh](#).

## Remarks

The high threshold is an absolute distance from the neutral surface (amplitude = 0). The high threshold extends further away from the neutral surface than the low threshold.

For positive [amplitude](#), the high threshold flattens the peaks of the surface finish. For negative amplitude, the high threshold flattens the valleys of the surface finish.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

## BumpRoughLow Property (IRenderMaterial)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpRoughLow Property (IRenderMaterial)

Gets or sets the low threshold of the surface finish for this appearance.

### .NET Syntax

#### Visual Basic (Declaration)

Property BumpRoughLow As System.Double

#### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpRoughLow = value
```

```
value = instance.BumpRoughLow
```

#### C#

```
System.double BumpRoughLow {get; set;}
```

#### C++/CLI

```
property System.double BumpRoughLow {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Low threshold of the surface finish

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpRoughLow](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## BumpScale Property (IRenderMaterial)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpScale Property (IRenderMaterial)

Gets or sets the scale for the surface-finish pattern incidence for this appearance.

### .NET Syntax

#### Visual Basic (Declaration)

Property BumpScale As System.Double

#### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpScale = value
```

```
value = instance.BumpScale
```

C#

```
System.double BumpScale {get; set;}
```

C++/CLI

```
property System.double BumpScale {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Scale

#### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.BumpScale](#).

#### Remarks

Scale controls the pattern of surface finish elements. A higher scale decreases the pattern incidence, and a lower scale increases the incidence of pattern elements.

Scale is more pronounced for low [amplitude](#) values. [Radius](#) also affects the pattern incidence for dimpled surface finishes.

#### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

#### Availability

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## BumpSharpness Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpSharpness Property (IRenderMaterial)

Gets or sets the sharpness, which influences the shape of the surface finish, of this appearance.

#### .NET Syntax

Visual Basic (Declaration)

```
Property BumpSharpness As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.BumpSharpness = value
```

```
value = instance.BumpSharpness
```

C#

```
System.double BumpSharpness {get; set;}
```

C++/CLI

```
property System.double BumpSharpness {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Sharpness

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial:BumpSharpness](#).

### Remarks

A higher sharpness value retains the original shape of the surface finish; a lower sharpness value filters (smoothenes) the surface-finish details.

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## BumpStrength Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpStrength Property (IRenderMaterial)

Gets or sets the amplitude of the surface layer for this appearance.

### .NET Syntax

Visual Basic (Declaration)

```
Property BumpStrength As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double  
  
instance.BumpStrength = value  
  
value = instance.BumpStrength
```

C#

```
System.double BumpStrength {get; set;}
```

C++/CLI

```
property System.double BumpStrength {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Amplitude

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpStrength](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2025 FCS, Revision Number 33

SOLIDWORKS API Help

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## BumpTextureFilename Property (IRenderMaterial)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpTextureFilename Property (IRenderMaterial)

Obsolete. Superseded by IRenderMaterial::SurfaceFinishFile. Gets or sets the path and file name of the pattern based on an image file for the surface finish of this appearance.

## .NET Syntax

Visual Basic (Declaration)

Property BumpTextureFilename As System.String

Visual Basic (Usage)



```
Dim instance As IRenderMaterial
Dim value As System.String

instance.BumpTextureFilename = value
value = instance.BumpTextureFilename
```

C#

```
System.string BumpTextureFilename {get; set;}
```

C++/CLI

```
property System.String^ BumpTextureFilename {
    System.String^ get();
    void set ( & System.String^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Path and file name of the pattern based on an image file for the surface finish

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpTextureFilename](#).

## Example

[Get Surface-finish Image Path and File Name \(C#\)](#)  
[Get Surface-finish Image Path and File Name \(VB.NET\)](#)  
[Get Surface-finish Image Path and File Name \(VBA\)](#)

## Remarks

This property corresponds to selecting **From File** as the surface finish type on the Surface Finish tab of the Appearances PropertyManager page and browsing to an image.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## BumpUseMappingScale Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : BumpUseMappingScale Property (IRenderMaterial)

Gets or sets whether to use the material's scale and mapping for the surface finish of this appearance.

## .NET Syntax

### Visual Basic (Declaration)

Property BumpUseMappingScale As System.Boolean

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Boolean  
  
instance.BumpUseMappingScale = value  
value = instance.BumpUseMappingScale
```

### C#

```
System.bool BumpUseMappingScale {get; set;}
```

### C++/CLI

```
property System.bool BumpUseMappingScale {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to use the material's scale and mapping, false to not

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::BumpUseMappingScale](#).

## Remarks

If BumpUseMappingScale is set to FALSE, then specify the [scale](#) and mapping for this surface finish for this appearance.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## CausticsCast Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : CausticsCast Property (IRenderMaterial)

Gets or sets whether specular materials reflect caustic photons for illuminating this appearance.

## .NET Syntax

Visual Basic (Declaration)

Property CausticsCast As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Boolean
```

```
instance.CausticsCast = value
```

```
value = instance.CausticsCast
```

C#

```
System.bool CausticsCast {get; set;}
```

C++/CLI

```
property System.bool CausticsCast {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True if specular materials reflect caustic photons, false if not

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.CausticsCast](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## CausticsReceive Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : CausticsReceive Property (IRenderMaterial)

Gets or sets whether diffuse materials absorb caustic photons for illuminating this appearance.

## .NET Syntax

Visual Basic (Declaration)

Property CausticsReceive As System.Boolean

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Boolean
```

```
instance.CausticsReceive = value  
value = instance.CausticsReceive
```

C#

```
System.bool CausticsReceive {get; set;}
```

C++/CLI

```
property System.bool CausticsReceive {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True if diffuse materials absorb caustic photons, false if not

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::CausticsReceive](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## ColorForm Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : ColorForm Property (IRenderMaterial)

Gets or sets the number of colors required to describe the appearance .

### .NET Syntax

Visual Basic (Declaration)

```
Property ColorForm As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Integer  
  
instance.ColorForm = value  
value = instance.ColorForm
```

C#

```
System.int ColorForm {get; set;}
```

C++/CLI

```
property System.int ColorForm {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Number of colors as defined by [swRenderMaterialColorForms\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.ColorForm](#).

## Remarks

For example, blue glass is defined by two colors. To see the corresponding user-interface controls, open a model to which the blue glass appearance has been applied. Edit the blue glass appearance to open the Appearances PropertyManager page. Two color controls, Current Color1 and Current Color2, are displayed under Color.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## CustomHoleMeshFile Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : CustomHoleMeshFile Property (IRenderMaterial)

Gets or sets the custom hole mesh file.

## .NET Syntax

Visual Basic (Declaration)

```
Property CustomHoleMeshFile As System.String
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.String  
  
instance.CustomHoleMeshFile = value  
value = instance.CustomHoleMeshFile
```

C#

```
System.string CustomHoleMeshFile {get; set;}
```

C++/CLI

```
property System.String^ CustomHoleMeshFile {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Path to custom hole mesh file

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.CustomHoleMeshFile](#).

## Remarks

This property is valid only if [IRenderMaterial.SurfaceFinishShaderType](#) is set to [swAppearanceSurfaceFinishShaderTypes\\_e.swAppearanceSurfaceFinishShaderType\\_CustomHoleMesh](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2025 FCS, Revision Number 33

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## DensityOfHoles Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : DensityOfHoles Property (IRenderMaterial)

Gets or sets the density of the holes of the mesh in corroded or eroded materials for illuminating this appearance.

## .NET Syntax

### Visual Basic (Declaration)

Property DensityOfHoles As System.Double

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.DensityOfHoles = value
```

```
value = instance.DensityOfHoles
```

### C#

```
System.double DensityOfHoles {get; set;}
```

### C++/CLI

```
property System.double DensityOfHoles {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Density of the holes of the mesh in corroded or eroded materials

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::DensityOfHoles](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## Diffuse Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : Diffuse Property (IRenderMaterial)

Gets or sets the intensity of a light source illuminating a surface from all directions without attenuation or shadowing for this appearance.

## Visual Basic (Declaration)

Property Diffuse As System.Double

## Visual Basic (Usage)

```
Dim instance As IRenderMaterial
Dim value As System.Double
```

```
instance.Diffuse = value
```

```
value = instance.Diffuse
```

## C#

```
System.double Diffuse {get; set;}
```

## C++/CLI

```
property System.double Diffuse {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Intensity of a light source illuminating a surface from all directions without attenuation or shadowing

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::Diffuse](#).

## Remarks

The appearance should contain an ambient light.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## DimpleSize Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : DimpleSize Property (IRenderMaterial)

Gets or sets the dimple size.

## Visual Basic (Declaration)

Property DimpleSize As System.Double

## Visual Basic (Usage)

```
Dim instance As IRenderMaterial
Dim value As System.Double
```

```
instance.DimpleSize = value  
value = instance.DimpleSize
```

C#

```
System.double DimpleSize {get; set;}
```

C++/CLI

```
property System.double DimpleSize {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Dimple size

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::DimpleSize](#).

### Remarks

This property is valid only if [IRenderMaterial::SurfaceFinishShaderType](#) is set to [swAppearanceSurfaceFinishShaderTypes\\_e.swAppearanceSurfaceFinishShaderType\\_Dimpled](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

SOLIDWORKS 2025 FCS, Revision Number 33

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## Direction1RotationAngle Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : Direction1RotationAngle Property (IRenderMaterial)

Gets or sets the angle at which to rotate the texture relative to the horizontal axis for mapping this appearance.

### .NET Syntax

Visual Basic (Declaration)

```
Property Direction1RotationAngle As System.Double
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.Direction1RotationAngle = value  
value = instance.Direction1RotationAngle
```

C#

```
System.double Direction1RotationAngle {get; set;}
```

C++/CLI

```
property System.double Direction1RotationAngle {  
    System.double get();
```

```
void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Angle at which to rotate the texture to the horizontal axis

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::Direction1RotationAngle](#).

### Remarks

This property affects spheres only.

Call [IRenderMaterial::Direction2RotationAngle](#) to get or set the angle to rotate the texture to the vertical axis.

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## Direction2RotationAngle Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : Direction2RotationAngle Property (IRenderMaterial)

Gets or sets the angle at which to rotate the texture relative to the vertical axis for mapping this appearance.

### .NET Syntax

#### Visual Basic (Declaration)

```
Property Direction2RotationAngle As System.Double
```

#### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double  
  
instance.Direction2RotationAngle = value  
value = instance.Direction2RotationAngle
```

#### C#

```
System.double Direction2RotationAngle {get; set;}
```

#### C++/CLI

```
property System.double Direction2RotationAngle {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Angle at which to rotate the texture relative to the vertical axis

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::Direction2RotationAngle](#).

## Remarks

This property affects spheres only.

Call [IRenderMaterial::Direction1RotationAngle](#) to get or set the angle to rotate the texture to the horizontal axis.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## DoubleSided Property (IRenderMaterial)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : DoubleSided Property (IRenderMaterial)

Gets or sets whether to enable shading from both sides of a surface when rendering a model using a ray-trace rendering engine.

## .NET Syntax

### Visual Basic (Declaration)

```
Property DoubleSided As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Boolean
```

```
instance.DoubleSided = value
```

```
value = instance.DoubleSided
```

### C#

```
System.bool DoubleSided {get; set;}
```

### C++/CLI

```
property System.bool DoubleSided {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to enable shading from both sides, false to not

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::DoubleSided](#).

## Example

[Get Surface-finish Image Path and File Name \(C#\)](#)  
[Get Surface-finish Image Path and File Name \(VB.NET\)](#)  
[Get Surface-finish Image Path and File Name \(VBA\)](#)

## Remarks

This method is only available when a ray-trace rendering engine is loaded.

## NOTES:

- When this property is disabled, surfaces that do not face the camera seem invisible.
- Double-sided surfaces can cause rendering errors. Use sparingly.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2016 FCS, Revision Number 24.0

SOLIDWORKS API Help

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## Emission Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : Emission Property (IRenderMaterial)

Gets or sets how much light is projected from the appearance.

## .NET Syntax

Visual Basic (Declaration)

Property Emission As System.Double

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.Emission = value
```

```
value = instance.Emission
```

C#

```
System.double Emission {get; set;}
```

C++/CLI

```
property System.double Emission {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Amount of light projected from the appearance

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::Emission](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## FaceWidth Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : FaceWidth Property (IRenderMaterial)

Gets or sets the face width.

## .NET Syntax

### Visual Basic (Declaration)

Property FaceWidth As System.Double

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Double
```

```
instance.FaceWidth = value
```

```
value = instance.FaceWidth
```

### C#

```
System.double FaceWidth {get; set;}
```

### C++/CLI

```
property System.double FaceWidth {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Face width

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.FaceWidth](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2025 FCS, Revision Number 33

SOLIDWORKS API Help

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## FileName Property (IRenderMaterial)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : FileName Property (IRenderMaterial)

Gets or sets the path and file name (.p2m) of the appearance.

## .NET Syntax

### Visual Basic (Declaration)

Property FileName As System.String

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.String
```

```
instance.FileName = value
```

```
value = instance.FileName
```

C#

```
System.string FileName {get; set;}
```

C++/CLI

```
property System.String^ FileName {  
    System.String^ get();  
    void set ( & System.String^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Path and file name (.p2m) of the appearance

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial.FileName](#).

### Example

[Add Decal \(VBA\)](#)  
[Add Decal \(VB.NET\)](#)  
[Add Decal \(C#\)](#)  
[Get Appearance File Name \(C#\)](#)  
[Get Appearance File Name \(VB.NET\)](#)  
[Get Appearance File Name \(VBA\)](#)

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## FitHeight Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : FitHeight Property (IRenderMaterial)

Gets or sets whether to stretch the height of the appearance texture to the selected entity when mapping the appearance.

### .NET Syntax

Visual Basic (Declaration)

```
Property FitHeight As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Boolean
```

```
instance.FitHeight = value
```

```
value = instance.FitHeight
```

C#

```
System.bool FitHeight {get; set;}
```

C++/CLI

```
property System.bool FitHeight {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to stretch the height of the appearance texture to the selected entity, false to not

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::FitHeight](#).

### Example

See the "Add Decal" examples in [IRenderMaterial::FileName](#).

### Remarks

Call [IRenderMaterial::FitWidth](#) to get or set whether to stretch the width of the appearance texture to the selected entity.

To stretch width and height of the appearance texture yourself, call [IRenderMaterial::Width](#) and [IRenderMaterial::Height](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

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## FitWidth Property (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : FitWidth Property (IRenderMaterial)

Gets or sets whether to stretch the width of the appearance texture to the selected entity when mapping the appearance.

### .NET Syntax

#### Visual Basic (Declaration)

```
Property FitWidth As System.Boolean
```

#### Visual Basic (Usage)



```
Dim instance As IRenderMaterial
Dim value As System.Boolean
```

```
instance.FitWidth = value
```

```
value = instance.FitWidth
```

#### C#

```
System.bool FitWidth {get; set;}
```

#### C++/CLI

```
property System.bool FitWidth {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to stretch the width of the appearance texture to the selected entity, false to not

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::FitWidth](#).

### Example

See the "Add Decal" examples in [IRenderMaterial::FileName](#).

### Remarks

Call [IRenderMaterial::FitHeight](#) to get or set whether to stretch the height of the appearance texture to the selected entity.

To stretch width and height of the appearance texture yourself, call [IRenderMaterial::Width](#) and [IRenderMaterial::Height](#).

### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

### Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## FixedAspectRatio Property (IRenderMaterial)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : FixedAspectRatio Property (IRenderMaterial)

Gets or sets whether the aspect ratio is fixed for mapping this appearance texture.

### .NET Syntax

#### Visual Basic (Declaration)

Property FixedAspectRatio As System.Boolean

#### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Boolean
```

```
instance.FixedAspectRatio = value
```

```
value = instance.FixedAspectRatio
```

#### C#

```
System.bool FixedAspectRatio {get; set;}
```

#### C++/CLI

```
property System.bool FixedAspectRatio {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True if the aspect ratio is fixed, false if not

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::FixedAspectRatio](#).

## Example

[Get Surface-finish Image Path and File Name \(C#\)](#)  
[Get Surface-finish Image Path and File Name \(VB.NET\)](#)  
[Get Surface-finish Image Path and File Name \(VBA\)](#)

## Remarks

When FixedAspect is true, then the image is uniformly scaled when either the [height](#) or [width](#) is changed.

Use this property when changing the height or width of an appearance texture and wanting it uniformly scaled.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## GetCenterPoint Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetCenterPoint Method (IRenderMaterial)

Obsolete. Superseded by [IRenderMaterial::GetCenterPoint2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Sub GetCenterPoint( _  
    ByRef CenterPt As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim CenterPt As System.Double  
  
instance.GetCenterPoint(CenterPt)
```

C#

```
void GetCenterPoint(  
    out System.double CenterPt  
)
```

C++/CLI

```
void GetCenterPoint(  
    & [Out] System.double CenterPt  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*CenterPt*  
Array of doubles (x, y, z) representing the centerpoint of the mapping for texture-based appearances

### Return Value

Array of doubles (x, y, z) representing the centerpoint of the mapping for texture-based appearances

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetCenterPoint](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::SetCenterPoint Method](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

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## GetCenterPoint2 Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetCenterPoint2 Method (IRenderMaterial)

Gets the center point of the mapping for texture-based appearances.

## .NET Syntax

### Visual Basic (Declaration)

```
Sub GetCenterPoint2( _  
    ByRef CenterPt\_X As System.Double, _  
    ByRef CenterPt\_Y As System.Double, _  
    ByRef CenterPt\_Z As System.Double _  
)
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim CenterPt_X As System.Double  
Dim CenterPt_Y As System.Double  
Dim CenterPt_Z As System.Double  
  
instance.GetCenterPoint2(CenterPt_X, CenterPt_Y, CenterPt_Z)
```

### C#

```
void GetCenterPoint2(  
    out System.double CenterPt\_X,  
    out System.double CenterPt\_Y,  
    out System.double CenterPt\_Z  
)
```

### C++/CLI

```
void GetCenterPoint2(  
& [Out] System.double CenterPt\_X,  
& [Out] System.double CenterPt\_Y,  
& [Out] System.double CenterPt\_Z  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

[CenterPt\\_X](#)  
X coordinate of the center point  
[CenterPt\\_Y](#)  
Y coordinate of the center point  
[CenterPt\\_Z](#)  
Z coordinate of the center point

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetCenterPoint2](#).

## Example

See [IRenderMaterial](#) examples.

 See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::SetCenterPoint2 Method \(\)](#)

 Availability

SOLIDWORKS 2013 SP05, Revision 21.5

SOLIDWORKS API Help

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GetDSPBRMaterial Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetDSPBRMaterial Method (IRenderMaterial)

Gets the DSPBRMaterial object.

 .NET Syntax

Visual Basic (Declaration)

```
Function GetDSPBRMaterial() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Object  
  
value = instance.GetDSPBRMaterial()
```

C#

```
System.object GetDSPBRMaterial()
```

C++/CLI

```
System.Object^ GetDSPBRMaterial();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

[IDSPBRMaterial](#)

 Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetDSPBRMaterial](#).

 Remarks

This method is valid only if [IRenderMaterial::HasDSPBRProperties](#) is true.

 See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)

 Availability

SOLIDWORKS 2026 FCS, Revision Number 34

SOLIDWORKS API Help

[Send comments on this topic.](#)

GetEntities Method (IRenderMaterial)

[See Also](#)



Gets the entities on which this appearance is applied.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetEntities() As System.Object
```

### Visual Basic (Usage)

```
Dim instance As IRenderMaterial  
Dim value As System.Object  
  
value = instance.GetEntities()
```

### C#

```
System.object GetEntities()
```

### C++/CLI

```
System.Object^ GetEntities();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of entities on which this appearance is applied

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetEntities](#).

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::IGetEntities Method](#)  
[IRenderMaterial::AddEntity Method](#)  
[IRenderMaterial::GetEntitiesCount Method](#)  
[IRenderMaterial::RemoveAllEntities Method](#)

## Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetEntitiesCount Method (IRenderMaterial)

[See Also](#)



SolidWorks.Interop.sldworks Namespace > IRenderMaterial Interface : GetEntitiesCount Method (IRenderMaterial)

Gets the number of entities on which this appearance was applied.

## .NET Syntax

### Visual Basic (Declaration)

```
Function GetEntitiesCount() As System.Integer
```

### Visual Basic (Usage)

```
Dim instance As IRenderMaterial  
Dim value As System.Integer  
  
value = instance.GetEntitiesCount()
```

### C#

```
System.int GetEntitiesCount()
```

C++/CLI

```
System.int GetEntitiesCount();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Number of entities on which this appearance was applied

#### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetEntitiesCount](#).

#### Remarks

Call this method before calling [IRenderMaterial::IGetEntities](#) to get the size of the array for that method.

#### See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::AddEntity Method](#)  
[IRenderMaterial::GetEntities Method](#)  
[IRenderMaterial::RemoveAllEntities Method](#)

#### Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GetLinkedDisplayStates Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetLinkedDisplayStates Method (IRenderMaterial)

Gets the display states to which this appearance is applied.

#### .NET Syntax

Visual Basic (Declaration)

```
Function GetLinkedDisplayStates() As System.Object
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim value As System.Object  
  
value = instance.GetLinkedDisplayStates()
```

C#

```
System.object GetLinkedDisplayStates()
```

C++/CLI

```
System.Object^ GetLinkedDisplayStates();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

Array of display state names

#### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetLinkedDisplayStates](#).

## Example

See the [IRenderMaterial](#) examples.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IModelDocExtension::SetLinkedDisplayStates Method \(\)](#)

## Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

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## GetMaterialIds Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetMaterialIds Method (IRenderMaterial)

Gets the material IDs of an appearance.

## .NET Syntax

### Visual Basic (Declaration)

```
Sub GetMaterialIds( _  
    ByRef PWMaterialId1 As System.Integer, _  
    ByRef PWMaterialId2 As System.Integer _  
)
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim PWMaterialId1 As System.Integer  
Dim PWMaterialId2 As System.Integer  
  
instance.GetMaterialIds(PWMaterialId1, PWMaterialId2)
```

### C#

```
void GetMaterialIds(  
    out System.int PWMaterialId1,  
    out System.int PWMaterialId2  
)
```

### C++/CLI

```
void GetMaterialIds(  
& [Out] System.int PWMaterialId1,  
& [Out] System.int PWMaterialId2  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*PWMaterialId1*  
First material ID  
*PWMaterialId2*  
Second material ID

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetMaterialIds](#).

## Example

See [IRenderMaterial](#) examples.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial Method](#)  
[IModelDocExtension::DeleteDisplayStateSpecificRenderMaterial Method](#)

## Availability

SOLIDWORKS 2011 FCS, Revision Number 19.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## GetUDirection Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetUDirection Method (IRenderMaterial)

Obsolete. Superseded by [IRenderMaterial::GetUDirection2](#).

## .NET Syntax

Visual Basic (Declaration)

```
Sub GetUDirection( _  
    ByRef UDir As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim UDir As System.Double  
  
instance.GetUDirection(UDir)
```

C#

```
void GetUDirection(  
    out System.double UDir  
)
```

C++/CLI

```
void GetUDirection(  
& [Out] System.double UDir  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*UDir*  
Array of doubles representing the U direction for the texture-based appearance (see **Remarks**)

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetUDirection](#).

## Remarks

Call [IRenderMaterial::GetVDirection](#) to get the V direction of the appearance.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::SetUDirection Method](#)  
[IRenderMaterial::SetVDirection Method](#)

## Availability

## GetUDirection2 Method (IRenderMaterial)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetUDirection2 Method (IRenderMaterial)

Gets the U direction of the texture-based appearance.

## .NET Syntax

## Visual Basic (Declaration)

```

Sub GetUDirection2( _
    ByRef UDir\_X As System.Double, _
    ByRef UDir\_Y As System.Double, _
    ByRef UDir\_Z As System.Double _
)

```

## Visual Basic (Usage)

```

Dim instance As IRenderMaterial
Dim UDir_X As System.Double
Dim UDir_Y As System.Double
Dim UDir_Z As System.Double

```

```
instance.GetUDirection2(UDir_X, UDir_Y, UDir_Z)
```

## C#

```

void GetUDirection2(
    out System.double UDir\_X,
    out System.double UDir\_Y,
    out System.double UDir\_Z
)

```

## C++/CLI

```

void GetUDirection2(
    & [Out] System.double UDir\_X,
    & [Out] System.double UDir\_Y,
    & [Out] System.double UDir\_Z
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Parameters

[UDir\\_X](#)  
X coordinate of the U direction

[UDir\\_Y](#)  
Y coordinate of the U direction

[UDir\\_Z](#)  
Z coordinate of the U direction

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetUDirection2](#).

## Example

See [IRenderMaterial](#) examples.

## Remarks

Call [IRenderMaterial::GetVDirection2](#) to get the V direction of the appearance.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::SetUDirection2 Method \(\)](#)  
[IRenderMaterial::SetVDirection2 Method \(\)](#)

## Availability

SOLIDWORKS 2013 SP05, Revision 21.5

SOLIDWORKS API Help

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## GetVDirection Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetVDirection Method (IRenderMaterial)

Obsolete. Superseded by [IRenderMaterial::GetVDirection2](#).

## .NET Syntax

### Visual Basic (Declaration)

```
Sub GetVDirection( _  
    ByRef VDir As System.Double _  
)
```

### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim VDir As System.Double  
  
instance.GetVDirection(VDir)
```

### C#

```
void GetVDirection(  
    out System.double VDir  
)
```

### C++/CLI

```
void GetVDirection(  
& [Out] System.double VDir  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Parameters

*VDir*

Array of doubles representing the V direction of the texture-based appearance (see **Remarks**)

## Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetVDirection](#).

## Example

The [mapping\\_type](#) (surface, projection, automatic, etc.) indirectly determines the V direction. This vector is perpendicular to the U direction.

Call [IRenderMaterial::GetUDirection](#) to get the U direction of the appearance.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::SetUDirection Method](#)  
[IRenderMaterial::SetVDirection Method](#)

## Availability

## GetVDirection2 Method (IRenderMaterial)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IRenderMaterial Interface](#) : GetVDirection2 Method (IRenderMaterial)

Gets the V direction of the texture-based appearance.

### .NET Syntax

#### Visual Basic (Declaration)

```
Sub GetVDirection2( _  
    ByRef VDir\_X As System.Double, _  
    ByRef VDir\_Y As System.Double, _  
    ByRef VDir\_Z As System.Double _  
)
```

#### Visual Basic (Usage)



```
Dim instance As IRenderMaterial  
Dim VDir_X As System.Double  
Dim VDir_Y As System.Double  
Dim VDir_Z As System.Double  
  
instance.GetVDirection2(VDir_X, VDir_Y, VDir_Z)
```

#### C#

```
void GetVDirection2(  
    out System.double VDir\_X,  
    out System.double VDir\_Y,  
    out System.double VDir\_Z  
)
```

#### C++/CLI

```
void GetVDirection2(  
& [Out] System.double VDir\_X,  
& [Out] System.double VDir\_Y,  
& [Out] System.double VDir\_Z  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

[VDir\\_X](#)  
X coordinate of the V direction  
[VDir\\_Y](#)  
Y coordinate of the V direction  
[VDir\\_Z](#)  
Z coordinate of the V direction

### Visual Basic for Applications (VBA) Syntax

See [RenderMaterial::GetVDirection2](#).

### Example

See [IRenderMaterial](#) examples.

### Remarks

The [mapping\\_type](#) (surface, projection, automatic, etc.) indirectly determines the V direction. This vector is perpendicular to the U direction.

Call [IRenderMaterial::GetUDirection2](#) to get the U direction of the appearance.

## See Also

[IRenderMaterial Interface](#)  
[IRenderMaterial Members](#)  
[IRenderMaterial::SetUDirection2 Method \(\)](#)  
[IRenderMaterial::SetVDirection2 Method \(\)](#)

## Availability

SOLIDWORKS 2013 SP05, Revision 21.5